AQ7.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Consider a function $f: V \times V \rightarrow R$ where $V \subseteq R^2$ defined by $f(v, w) = 2v_1 w_1 + 5v_2 w_2$, where $v = (v_1, v_2), w = (w_1, w_2)$. Choose the set of correct options.

- [ ] f satisfies the symmetry condition of the inner product.
- [ ] f satisfies the bilinearity condition of the inner product.
- [ ] f satisfies the positivity condition of the inner product.
- [ ] f is an inner product.
- [ ] f is not an inner product.

No, the answer is incorrect.
Score: 0
Accepted Answers:
f satisfies the symmetry condition of the inner product.
f satisfies the bilinearity condition of the inner product.
f satisfies the positivity condition of the inner product.
f is an inner product.

1 point
Consider a function $f: V \times V \rightarrow R$ where $V \subseteq R^2$ defined by $f(v, w) = v_1 w_1 - v_1 w_2 - v_2 w_1 + 4v_2 w_2$, where $v = (v_1, v_2), w = (w_1, w_2)$. Choose the set of correct options.

- [ ] f satisfies the symmetry condition of the inner product.
- [ ] f satisfies the bilinearity condition of the inner product.
- [ ] f satisfies the positivity condition of the inner product.
- [ ] f is an inner product.
- [ ] f is not an inner product.

No, the answer is incorrect.
Score: 0
Accepted Answers:
f satisfies the symmetry condition of the inner product.
f satisfies the bilinearity condition of the inner product.
f satisfies the positivity condition of the inner product.
f is an inner product.

1 point

Consider two vectors $a = (0.4, 1.3, -2.2), b = (2, 3, -5)$ in R^3. Choose the set of correct options.

☐ The two vectors satisfy the triangle inequality given by $\| a + b \| \leq \| a \| + \| b \|$.

☐ The two vectors do not satisfy the triangle inequality.

☐ The two vectors satisfy the Cauchy-Schwarz inequality given by $| < a, b > | \leq \| a \| \| b \|$.

☐ The two vectors do not satisfy the Cauchy-Schwarz inequality.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The two vectors satisfy the triangle inequality given by $\| a + b \| \leq \| a \| + \| b \|$.

The two vectors satisfy the Cauchy-Schwarz inequality given by $| < a, b > | \leq \| a \| \| b \|$.

Level 2

1 point

Consider a function $f : V \times V \rightarrow R$ where $V \subseteq R^2$ defined by $f(v, w) = v_1^2 w_1^2 + v_1 w_2^2 + v_2^2 w_1$, where $v = (v_1, v_2), w = (w_1, w_2)$. Choose the set of correct options.

☐ f satisfies the symmetry condition of the inner product.

☐ f satisfies the positivity condition of the inner product.

☐ f is an inner product.

☐ f is not an inner product.

No, the answer is incorrect.
Score: 0
Accepted Answers:

f satisfies the symmetry condition of the inner product.

f is not an inner product.

1 point

Consider a vector $a = (a_1, b_1, c_1)$ in R^3. Which of these is (are) possible candidates for a norm?

☐ $\sqrt{a_1^2 + b_1^2 + c_1^2}$

☐ $a_1 - b_1$

☐

$a_1 + b_1 + c_1$ a1 + b1 + c1

☐

$max\{a_1, b_1, c_1\}$max{a1, b1, c1}

☐

$min\{a_1, b_1, c_1\}$min{a1, b1, c1}

☐

$\max\{|a_1|, |b_1|, |c_1|\}$max{|a1|, |b1|, |c1|}

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\sqrt{a_1^2 + b_1^2 + c_1^2}$ a12 + b12 + c12

√

$\max\{|a_1|, |b_1|, |c_1|\}$max{|a1|, |b1|, |c1|}

1 point

Choose the set of correct statements.

☐

$V = R^2$ V = R2 and the function $< .,. > : V \times V \longrightarrow R$< .,. >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > = $5x_1y_1 + 8x_2y_2 - 6x_1y_2 - 6x_2y_1$ 5x1y1 + 8x2y2 − 6x1y2 − 6x2y1 is an inner product on VV .

☐

$V = R^2$ V = R2 and the function $< .,. > : V \times V \longrightarrow R$< .,. >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > = $|x_1| + 2|y_2|$ |x1| + 2|y2| is an inner product on V V .

☐

$V = R^2$ V = R2 and the function $< .,. > : V \times V \longrightarrow R$< .,. >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > =$3x_1y_1 + 5x_2y_2$ 3x1y1 + 5x2y2 is an inner product on VV .

☐

$V = R^2$ V = R2 and the function $< .,. > : V \times V \longrightarrow R$< .,. >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > =$3x_1y_1 - 5x_2y_2$ 3x1y1 − 5x2y2 is an inner product on VV .

No, the answer is incorrect.
Score: 0
Accepted Answers:

$V = R^2$ V = R2 and the function $< .,. > : V \times V \longrightarrow R$< .,. >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > = $5x_1y_1 + 8x_2y_2 - 6x_1y_2 - 6x_2y_1$ 5x1y1 + 8x2y2 − 6x1y2 − 6x2y1 is an inner product on VV .

$V = R^2$ V = R2 and the function $< .,. > : V \times V \longrightarrow R$< .,. >: V × V ⟶ R defined as $< (x_1, x_2), (y_1, y_2) >$ < (x1, x2), (y1, y2) > =$3x_1y_1 + 5x_2y_2$ 3x1y1 + 5x2y2 is an inner product on VV .

Check Answers

Your score is: 0/6